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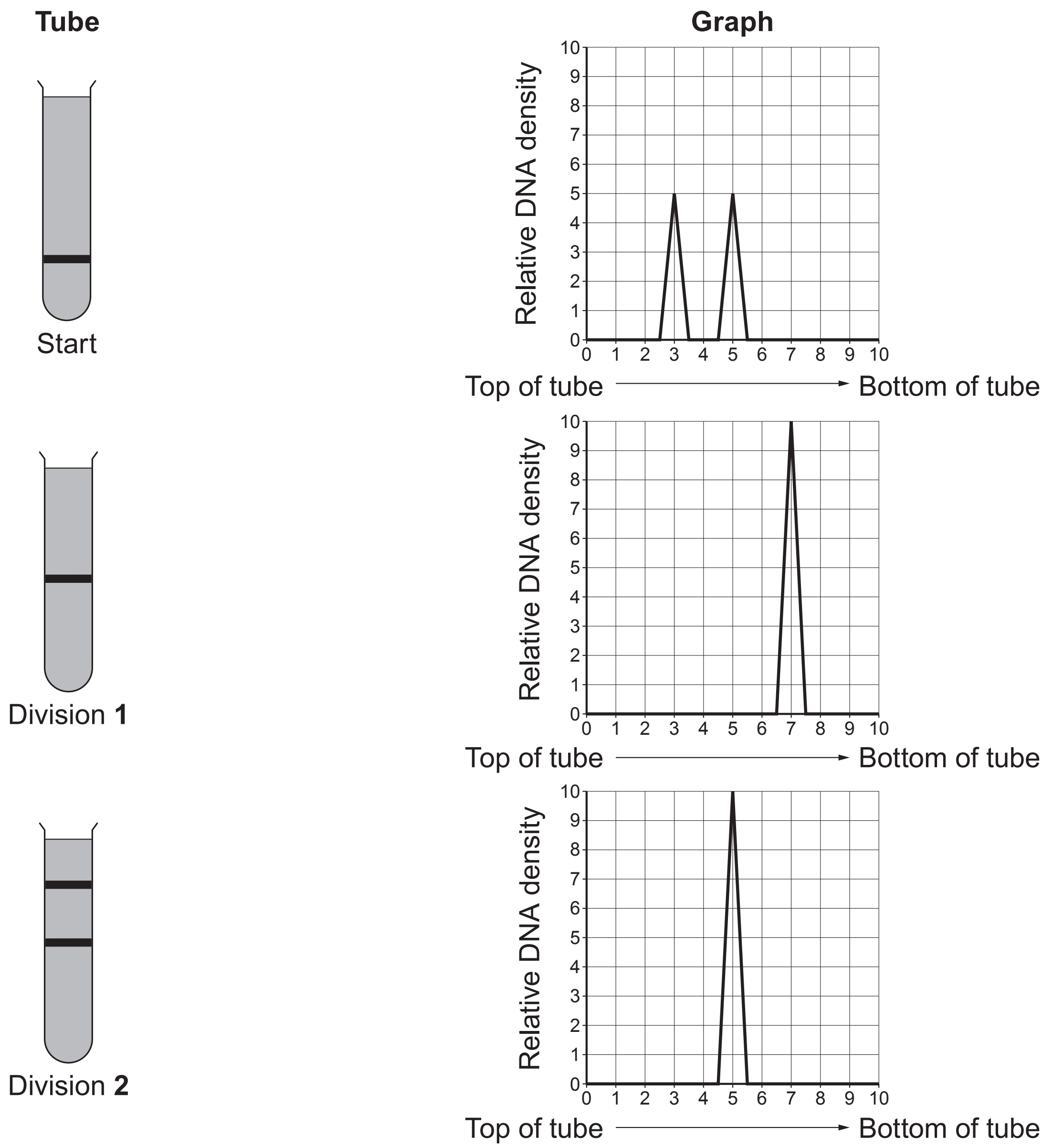
3. (a) In 1958 Meselson and Stahl conducted an investigation into DNA replication.

- Bacteria were grown in a culture medium that contained only the heavy isotope of nitrogen, ^{15}N. After many generations, the bacterial DNA contained only the heavy isotope of nitrogen.
- Some of the bacteria were then transferred to another culture medium containing only the lighter isotope of nitrogen, ^{14}N.
- DNA was extracted from the bacteria and centrifuged at the **start** (before cell division), and then after **1**, **2** and **3** cell divisions. Heavier DNA was found towards the bottom of the tube after centrifuging.

Image 3.1 shows the distribution of DNA in the tubes at the start and after Division **1** and Division **2**. The graphs represent the relative densities of DNA but they are not in the correct order.

(i) **Draw lines** on **Image 3.1** to match up the tubes with the correct graphs. [1]

Image 3.1



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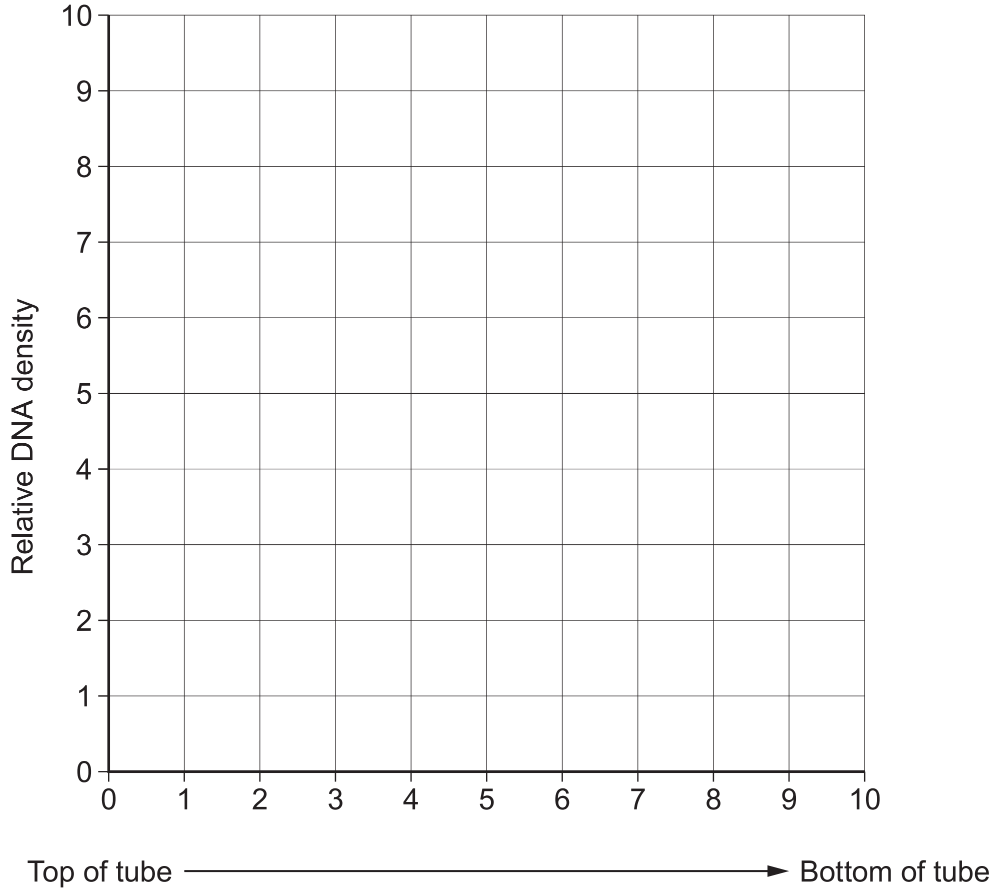
(ii) On **Image 3.2 sketch the graph** you would expect to see for Division **3**. [3]

Image 3.2



2400U101 09

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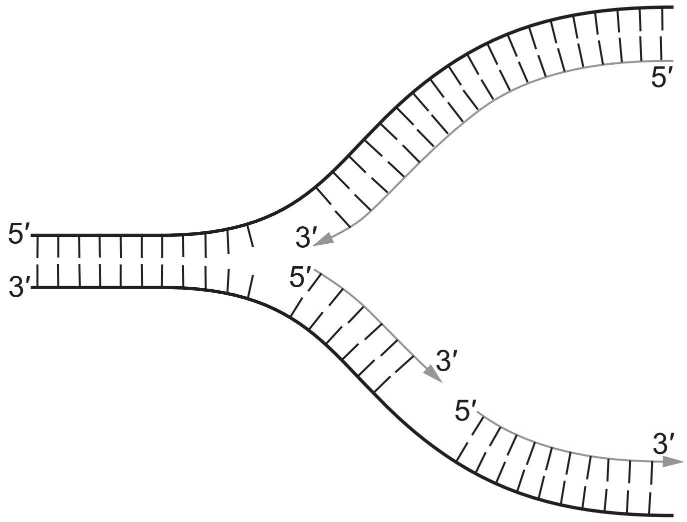
(b) **Image 3.3** shows a DNA replication fork.

Image 3.3



(i) On **Image 3.3**, mark the position of **one** molecule of DNA polymerase with the letter **X**. [1]

(ii) Describe the role of DNA helicase in this process. [1]

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(iii) Explain how **Image 3.3** shows that DNA is replicated by a semi-conservative mechanism. [2]

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(c) DNA polymerase has a proof-reading function, this means that it will check that it has replicated the DNA correctly and repair it if it is incorrectly synthesised.
Use your knowledge of protein synthesis to suggest the importance of the proof-reading role to the organism. [3]

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